

OPTIMAL DESIGN OF ACCELERATED LIFE TEST PLANS UNDER PERIODIC INSPECTION AND TYPE I CENSORING: THE CASE OF RAYLEIGH FAILURE LAW

Nesar Ahmad: Department of Statistics and Operations Research, Aligarh Muslim University, Aligarh, India.
A. Islam: Department of Statistics and Operations Research, Aligarh Muslim University, Aligarh, India.
R. Kumar: Department of Statistics, M.D. University, Rohtak, India.
R.K. Tuteja: Department of Statistics, M.D. University, Rohtak, India.

Key words: Accelerated life test; optimal design; periodic inspection; Rayleigh distribution.

Summary: Statistically optimal accelerated life test plans are determined for the case of Rayleigh failure distribution under the assumptions of periodic inspection and Type I censoring and two stress levels. For optimal plans the proportion of test units allocated to the low stress level and equally spaced inspection times are determined such that the asymptotic variance of the maximum likelihood estimator of p -th quantile at the use condition is minimized. Procedures for selecting a sample size are also discussed.

1. INTRODUCTION

Testing devices with high reliability under the usual operating conditions is difficult, because they are not likely to fail in the time available for tests. Accelerated life tests (ALT) quickly provide information on life distributions, and costs can be reduced by employing periodic inspection. Yum and Choi (1989) combined acceleration and periodic inspection and developed an asymptotically optimal ALT plan for exponentially distributed life times under the assumptions of Type I censoring and periodic inspection. The present investigation features the Rayleigh distribution of life times of test

units. Dyer and Whisenand (1973a & b) have assumed Rayleigh distribution as a model of life times of test units. They have also discussed various uses of the model in reliability and have found the distribution to be extremely important in communication engineering. For example, the envelope of a narrow-band Gaussian random process and the amplitude of atmospheric radio noise caused by the radiation due to lightning discharges in storms have probability density functions given by (1). An asymptotically optimal ALT plan is developed under Type I censoring and periodic inspection at two stress levels. Inspection times at each stress level are equally spaced between initial and censored times. The asymptotic variance of the estimated mean or p -th quantile of the life time distribution at the use condition is adopted as a criterion for determining optimal design.

2. THE MODEL

Let us assume that the life times (T) of test items at stress level s are independently and identically Rayleigh distributed with the probability density function

$$(1) \quad f(t) = \frac{t}{\sigma^2} e^{(-t^2/2\sigma^2)}, \quad t, \sigma > 0.$$

The mean life time is $\theta = \sqrt{\pi/2} \sigma$,

$$(2) \quad \text{where } \theta = e^{(\beta_0 + \beta_1 s)}$$

and β_0 and β_1 are constants to be estimated.

Suppose that s_0 is the operating stress level. Two test stress levels are considered, a low level s_1 and a high level s_2 . Levels s_0 and s_2 are prespecified, and s_1 ($s_1 < s_2$) is to be optimally determined. The number of test items allocated to s_1 and s_2 are given by

$$(3) \quad n_1 = \alpha_1 N, \quad n_2 = \alpha_2 N = (1 - \alpha_1)N,$$

where $\alpha_1 + \alpha_2 = 1$, N is the total number of test items given and α_1 is to be optimally determined.

At s_i , n_i units are to be put on test at time 0 and run until a pre-specified time t_{ci} (i.e. Type I censoring is assumed), and inspections are conducted only at specified points in time $t_{i1}, t_{i2}, \dots, t_{i,K(i)}$, where $t_{i,K(i)} = t_{ci}$.

In addition, let $t_{i0} = 0$ and $t_{i,K(i)+1} = \infty$, and at stress level s_i , we define

$$(4) \quad x_{ij} = \text{the number of items failed in } (t_{i,j-1}, t_{ij}),$$

and

$$P_{ij} = \text{the probability of failure in } (t_{i,j-1}, t_{ij})$$

$$(5) \quad = e^{-t_{i,j-1}^2 / 2\theta_i^2} - e^{-t_{ij}^2 / 2\theta_i^2}$$

for $j = 1, 2, \dots, K(i) + 1$, where $\theta_i = e^{(\beta_0 + \beta_1 s_i)}$.

The logarithm of mean life time at the use condition is given by

$$(6) \quad \mu_0 = \ln \theta_0 = \beta_0 + \beta_1 s_0.$$

We are interested in estimating the p -th quantile (t_p) of the Rayleigh distribution at the use condition, which is related to μ_0 as follows:

$$(7) \quad y_p = \ln t_p = \frac{1}{2} \ln 2 + \mu_0 + \frac{1}{2} \ln \{-\ln(1-p)\}.$$

Let $\hat{\beta}_0$ and $\hat{\beta}_1$ be estimators of β_0 and β_1 , respectively. Then y_p is estimated as

$$(8) \quad \hat{y}_p = \frac{1}{2} \ln 2 + \hat{\beta}_0 + \hat{\beta}_1 s_0 + \frac{1}{2} \ln \{-\ln(1-p)\}.$$

3. ESTIMATION OF PARAMETERS

The counts $\{x_{ij}, j = 1, 2, \dots, K(i) + 1\}$ are multinomially distributed with parameters n_i and $\{P_{ij}, j = 1, 2, \dots, K(i) + 1\}$, at stress level s_i . The log likelihood function is given by

$$(9) \quad L = C + \sum_{i=1}^2 \sum_{j=1}^{K(i)+1} x_{ij} \ln(P_{ij}),$$

where C is a constant.

Then, the ML estimates of β_0 and β_1 can be obtained by solving the following set of equations.

$$(10) \quad \frac{\partial L}{\partial \beta_r} = \sum_{i=1}^2 s_i^r \sum_{j=1}^{K(i)+1} (x_{ij}(B_{i,j-1} - B_{ij})/P_{ij}) = 0,$$

for $r = 0, 1$,

$$(11) \quad \text{where } B_{ij} = (t_{ij}/\theta_i)^2 e^{-(t_{ij}/\sqrt{2} \theta_i)^2}$$

The Fisher information matrix F , is given by

$$(12) \quad F = N(f_{gh}), \quad g, h = 0, 1,$$

where

$$(13) \quad f_{00} = \sum_{i=1}^2 \alpha_i Q_i, \quad f_{01} = f_{10} = \sum_{i=1}^2 \alpha_i s_i Q_i, \quad f_{11} = \sum_{i=1}^2 \alpha_i s_i^2 Q_i$$

and

$$(14) \quad Q_i = \sum_{j=1}^{K(i)+1} (B_{i,j-1} - B_{ij})^2 / P_{ij}$$

The asymptotic covariance matrix of the maximum likelihood estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ is the inverse of the corresponding Fisher information matrix (F^{-1}).

From (8) the asymptotic variance (avar) of $\hat{y}_p$ is obtained as

$$(15) \quad \text{avar}(\hat{y}_p) = \text{avar}(\hat{\beta}_0 + \hat{\beta}_1 s_0) = N^{-1} (f_{00} f_{11} - f_{01}^2)^{-1} (f_{11} + s_0^2 f_{00} - 2s_0 f_{01}).$$

The problem of optimally designing the ALT plan under periodic inspection can now be stated as: given $N, s_0, s_2, \{t_{ci}, i = 1, 2\}$ determine α_1, s_1 , and $\{t_{ij}, i = 1, 2; j = 1, 2, \dots, K(i)\}$ such that $\text{avar}(\hat{y}_p)$ is minimized.

4. STATISTICALLY OPTIMAL PLANS

We make the following simplifying assumptions:

- (1) Censoring times at s_1 and s_2 are the same. That is, $t_{c1} = t_{c2} = t_c$.
 (2) The number of inspections at each stress level is the same. That is, $K(1) = K(2) = K$, and K is assumed given.

Furthermore, parameters are standardized such that common censoring time (t_c) is 1, $s_0 = 0$, and $s_2 = 1$.

Based upon this standardization (15) becomes

$$(16) \quad \text{avar}(\hat{y}_p) = N^{-1}(f_{00}f_{11} - f_{01}^2)^{-1}f_{11}.$$

After some algebraic manipulation we obtain

$$(17) \quad \text{avar}(\hat{y}_p) = N^{-1}(Q_2 + (s_1^2 Q_1 - Q_2)\alpha_1)/(Q_1 Q_2 (s_1 - 1)^2 (-\alpha_1^2 + \alpha_1)).$$

Under the above assumptions, optimal plans are developed by determining optimal values of s_1 and α_1 for given N and K such that $\text{avar}(\hat{y}_p)$ is minimized. The optimization procedure is initiated by first 'guess estimate' of P_u and P_h which are more familiar and easier to estimate than β_0 and β_1 . That is,

$P_u = P_r$ {an item fails in $(0, t_c = 1)$ at the operating condition}

$P_h = P_r$ {an item fails in $(0, t_c = 1)$ at the high stress}.

Then, β_0 and β_1 are determined as follows:

$$(18) \quad \beta_0 = \frac{1}{2} \ln(-\frac{1}{2} \ln(1 - P_u)).$$

$$(19) \quad \beta_1 = \frac{1}{2} \ln\{\ln(1 - P_u)/\ln(1 - P_h)\}.$$

For given P_u and P_h , the minimum value of $\text{avar}(\hat{y}_p)$ with respect to s_1 and α_1 is determined by a two-step procedure, which was developed by Yum and Choi (1989).

When $K(1) = K(2) = K$ go to infinity (i.e. continuous inspection) then minimum $\text{avar}(\hat{y}_p)$ and corresponding optimal s_1 (i.e. s_1^*) and α_1 (i.e. α_1^*) were determined by using the method described in Nelson and Meeker's paper in 1978. It can also be determined by a two-step procedure. Finally, the ratio of $\text{avar}(\hat{y}_p(K))$ to $\text{avar}(\hat{y}_p(\infty))$ was determined.

5. RESULTS AND DISCUSSIONS

The results are summarized in tables 1 and 2 (more detailed computational results are available from the authors). Table 1 elaborates statistically optimal plans and table 2 shows sensitivity analysis of $\text{avar}(\hat{y}_p)$, for various combinations of P_u , P_h and K . We observe that $\text{avar}(\hat{y}_p)$ is not sensitive to K . Also, s_1^* and α_1^* are fairly stable over K for given P_u and P_h . It is clear that $\text{avar}(\hat{y}_p)$ for $K = 2, 5$ or 10 is close to that for $K = \infty$. This implies that the number of inspections (K) need not be too large. This is an achievement in life testing of an item which has high reliability. The ratio of $\text{avar}(\hat{y}_p(K))$ to $\text{avar}(\hat{y}_p(\infty))$ is taken in the last column of the table, and is approximately unity. This implies that there is no significant difference in the asymptotic variance of $\hat{y}_p$, which is an encouraging result in terms of testing efforts. Also, when P_u increases and P_h decreases, the value of s_1^* is close to the use stress and the value of α_1^* tends to 1. This shows that there is almost no need for an ALT. Similar trends are observed when the values of P_h are very small for all P_u .

One can determine the approximate sample size N such that

$$(20) \quad P_r\{\theta_0/h \leq \hat{\theta}_0 \leq h\theta_0\} \leq \phi,$$

where $h(>1)$ and ϕ are given constants.

Equation (20) can be rewritten as

$$(21) \quad P_r\{\hat{\mu}_0 - \ell n h \leq \hat{\mu}_0 \leq \hat{\mu}_0 + \ell n h\} \geq \phi.$$

The sample size (N) that approximately achieves this is:

$$(22) \quad N^* \cong v_0[(w/2)/\ell n h]^2,$$

where v_0 is the asymptotic variance of $\hat{y}_p$ when $N = 1$ and w is the $(1 + \phi)/2$ quantile of the standard normal distribution.

6. CONCLUSION

In this article, statistically optimal ALT plans are developed in which two stress levels are involved under the assumptions of periodic inspection at equally spaced time and type-I censoring. The Rayleigh distribution is considered to describe the failure of test units.

We conclude that, a large number of inspections (K) are not needed and estimated failure probabilities are fairly tolerable from their true values of the parameter considered. We also conclude that (1) the asymptotic variance of $\hat{y}_p$ is far less than that of the asymptotic variance obtained by Yum and Choi (1989) and (2) the overall trend is similar to the trend obtained by Yum and Choi (1989).

ACKNOWLEDGEMENTS

The authors are thankful to the referee and Professor J.S. Maritz, editor, for useful comments and constructive suggestions for improving the article.

REFERENCES

- DYER, D.D. & WHISENAND, C.W. 1973a. Best linear unbiased estimator of the parameter of the Rayleigh distribution - Part I: Small sample theory for censored order statistics. *IEEE Transactions on Reliability R.* 22, 27-34.
- DYER, D.D. AND WHISENAND, C.W. 1973b. Best linear unbiased estimator of the parameter of the Rayleigh distribution - Part II: Optimum theory for selected order statistics. *IEEE Transactions on Reliability, R.* 22, 229-231.
- MEEKER, W.Q. 1986. Planning life tests in which units are inspected for failure. *IEEE Transactions on reliability, R.* 35, 571-578.
- NELSON, W. & MEEKER, W.Q. 1978. Theory for optimum accelerated censored life tests for Weibull and extreme value distributions. *Technometrics*, 20, 171-177.
- YUM, B.J. & CHOI, S.C. 1989. Optimal design of accelerated life tests under periodic inspection. *Naval Research Logistics*, 36, 779-795.

TABLE 1. OPTIMAL ALT PLANS

P_u	P_h	K	δ_1^*	α_1^*	Navar($\hat{y}_p$)	RATIO
0,0001	0,9	2	0,787	0,708	41,0588	1,0283
		5	0,710	0,794	40,1125	1,0047
		10	0,710	0,795	39,9712	1,0011
		∞	0,710	0,796	39,9269	1,0000
	0,5	2	0,698	0,823	86,4579	1,0029
		5	0,698	0,824	86,2504	1,0005
		10	0,698	0,824	86,2179	1,0001
		∞	0,698	0,824	86,2076	1,0000
	0,1	2	0,630	0,849	317,2028	1,0001
		5	0,630	0,849	317,1865	1,0000
		10	0,630	0,849	317,1838	1,0000
		∞	0,630	0,849	317,1830	1,0000
	0,01	2	0,444	0,890	1318,6085	1,0000
		5	0,444	0,890	1318,6080	1,0000
		10	0,444	0,890	1318,6079	1,0000
		∞	0,444	0,890	1318,6079	1,0000
0,001	0,9	2	0,620	0,809	23,1310	1,0256
		5	0,624	0,814	22,6495	1,0042
		10	0,624	0,815	22,5773	1,0010
		∞	0,624	0,816	22,5547	1,0000
	0,5	2	0,590	0,847	44,7745	1,0025
		5	0,592	0,846	44,6812	1,0004
		10	0,592	0,846	44,6664	1,0001
		∞	0,592	0,846	44,6618	1,0000
	0,1	2	0,446	0,888	129,8203	1,0000
		5	0,446	0,888	129,8154	1,0000
		10	0,446	0,888	129,8145	1,0000
		∞	0,446	0,888	129,8143	1,0000
0,01	0,9	2	0,458	0,852	10,2933	1,0200
		5	0,464	0,855	10,1247	1,0033
		10	0,466	0,855	10,0993	1,0008
		∞	0,466	0,855	10,0913	1,0000
	0,5	2	0,368	0,898	16,6439	1,0017
		5	0,368	0,899	16,6205	1,0003
		10	0,368	0,899	16,6169	1,0001
		∞	0,368	0,899	16,6157	1,0000

TABLE 2. SENSITIVITIES OF $\text{avar}(\hat{y}_p)$

P_u	P_h				
0,0001	0,9	$\tilde{P}_u \setminus \tilde{P}_h$	0,8000	0,9000	0,9500
		0,00005	1,0149	1,0087	1,0058
		0,00010	1,0014	1,0000	1,0015
		0,00020	1,0087	1,0103	1,0160
	0,1	$\tilde{P}_u \setminus \tilde{P}_h$	0,0800	0,1000	0,1200
		0,00005	1,0072	1,0146	1,0226
		0,00010	1,0016	1,0000	1,0012
		0,00020	1,0340	1,0178	1,0101
0,001	0,9	$\tilde{P}_u \setminus \tilde{P}_h$	0,8000	0,9000	0,9500
		0,00050	1,0178	1,0152	1,0115
		0,00100	1,0011	1,0000	1,0009
		0,00200	1,0232	1,0195	1,0214
	0,1	$\tilde{P}_u \setminus \tilde{P}_h$	0,0800	0,1000	0,1200
		0,00050	1,0151	1,0302	1,0465
		0,00100	1,0043	1,0000	1,0024
		0,00200	1,0883	1,0455	1,0224
0,01	0,9	$\tilde{P}_u \setminus \tilde{P}_h$	0,8000	0,9000	0,9500
		0,00500	1,0245	1,0292	1,0302
		0,01000	1,0020	1,0000	1,0008
		0,02000	1,0678	1,0411	1,0332
	0,5	$\tilde{P}_u \setminus \tilde{P}_h$	0,4000	0,5000	0,6000
		0,00500	1,0184	1,0393	1,0616
		0,01000	1,0069	1,0000	1,0040
		0,02000	1,1338	1,0622	1,0265